



INTELLIGENT EV TRIP PLANNING & CHARGING INFRASTRUCTURE PLATFORM

CHARGEAHEAD

Don't just find a charger. Know you can charge.

Plan smarter. Charge confidently. Keep moving.

TEAM 6

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A charger marked "Available" can still leave an EV stranded

A REAL-WORLD JOURNEY



15 km from charger

Map shows: Available



Arrives at the station

Charger occupied · 3 vehicles waiting



Second charger down

Marked online, but malfunctioning



Payment app failing

No way to start a session

Current systems show infrastructure — not charging certainty.



Range

Can I actually reach it?



Availability

Will it be free on arrival?



Reliability

Will it actually work?



Queue

How long will I wait?

4 uncertainties stand between a driver and a confident charge.

EV navigation today is fragmented

TODAY	CHARGEAHEAD
Current charger status	✔ Predicted availability
Generic vehicle range	✔ Journey-specific range
No physical queue visibility	✔ Queue intelligence
Online ≠ functional	✔ Reliability score
Static route	✔ Live rerouting
Separate operator apps	✔ Unified charging experience

Maps answer WHERE. **ChargeAhead answers WHETHER.**

One intelligent platform for the entire EV charging journey



PREDICT

- Smart range awareness
- Predictive charger availability
- Queue prediction
- Station reliability



ACT

- Optimal charging stop
- Pre-booking
- Walk-in option
- Unified payment

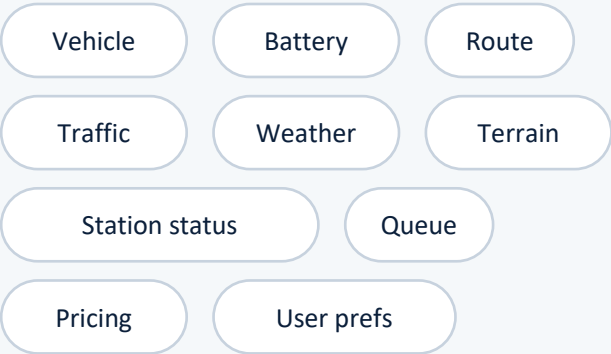


ADAPT

- Live station monitoring
- Community reports
- Failure detection
- Dynamic rerouting

We recommend the best charger for YOU — not the nearest one

INPUTS



OUTPUT

- ✓ Best charging stop
- ✓ Best arrival time
- ✓ Expected wait time
- ✓ Charging duration & cost
- ✓ Backup station

STATION A

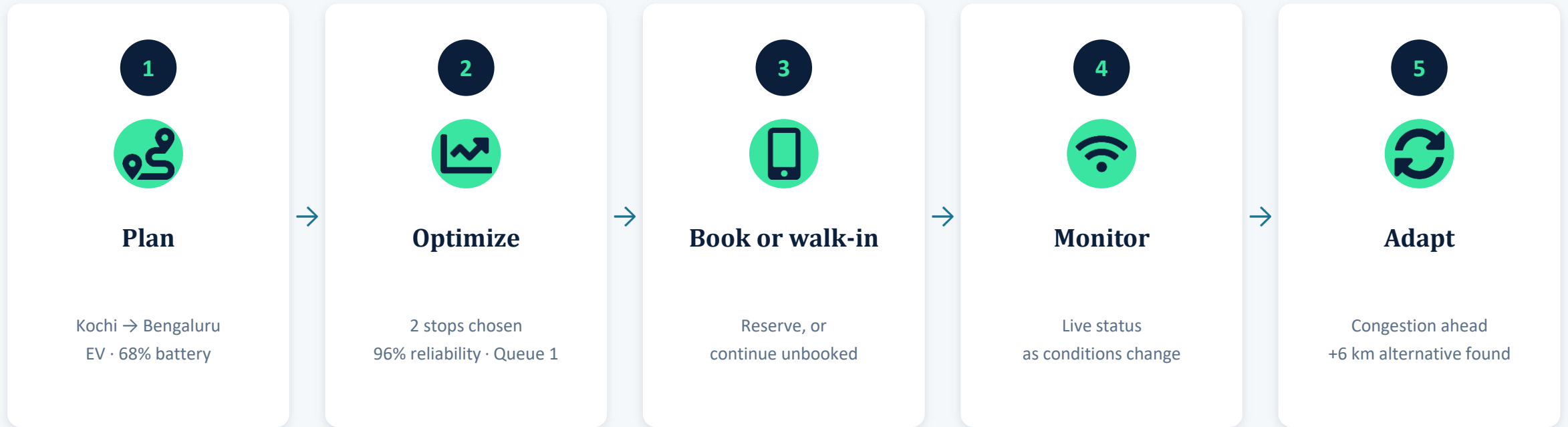
● Free now ETA 30 min Queue: 4

STATION B

● Busy now ETA 40 min Queue: 0

Recommendation:
Station B — 10 min faster overall

From destination to destination — without charging anxiety



ARRIVE WITH CONFIDENCE

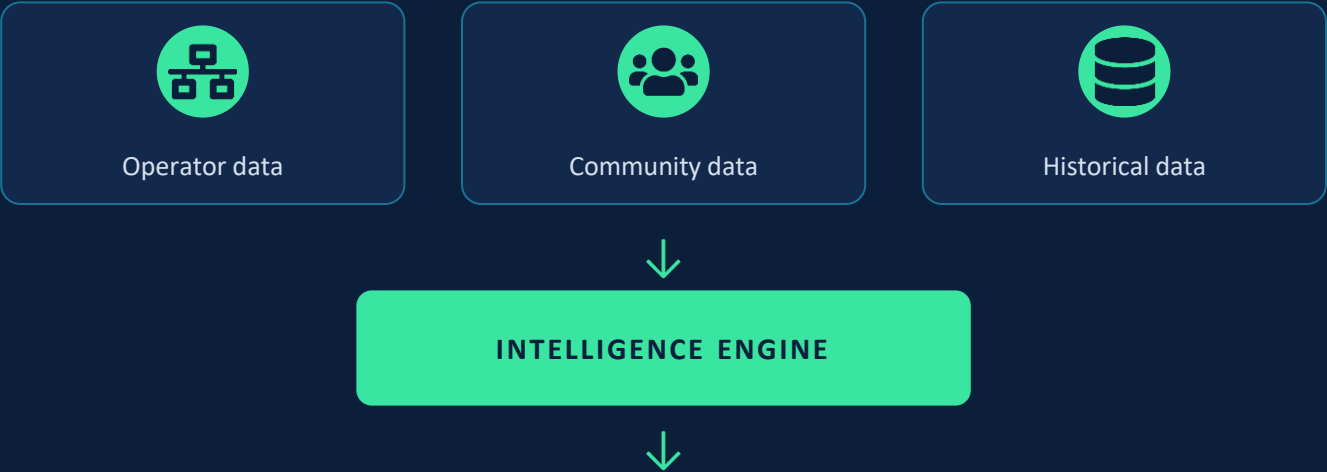
OUR DIFFERENTIATOR

True Charging Availability

Instead of simply showing:

AVAILABLE

ChargeAhead calculates a full confidence picture:



TRUE STATION STATUS — Availability + Queue + Reliability + ETA

STATION CONFIDENCE

92%

2 chargers available

Low predicted queue

Recently verified

High historical reliability

COMMUNITY LAYER — users can flag:

Broken

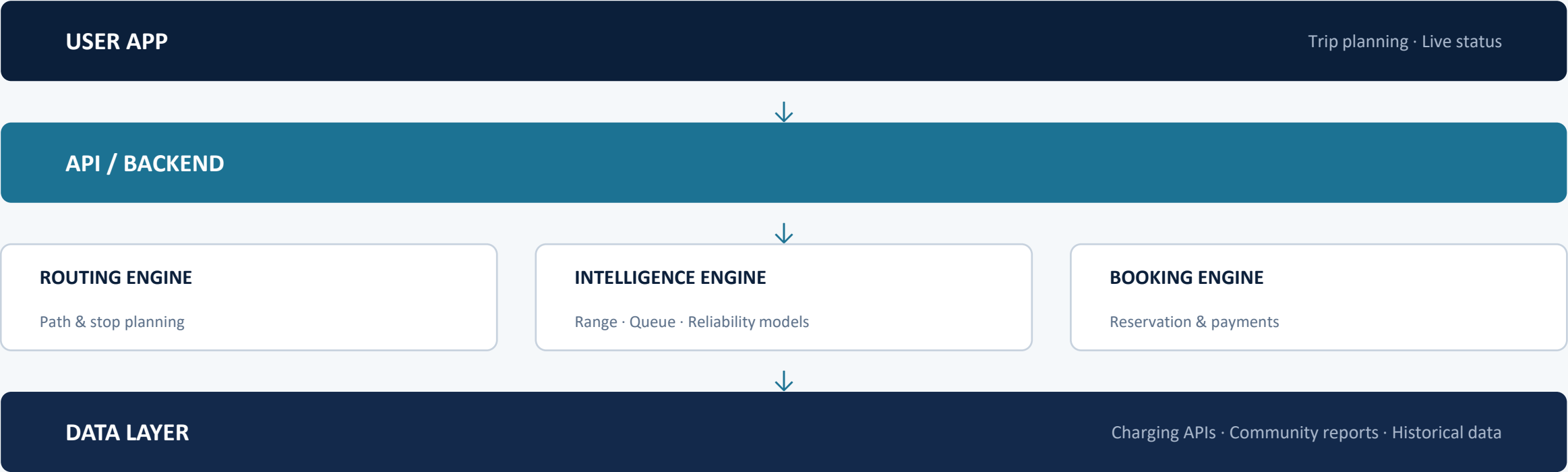
Blocked

Busy

Working

Payment issue

A scalable intelligence layer above charging infrastructure



SUGGESTED STACK

- Flutter / React Native
- FastAPI / Node.js
- PostgreSQL + PostGIS
- Python · XGBoost
- WebSockets / MQTT
- AWS / Azure / GCP

Built on existing infrastructure. Designed for a scalable business.

TECHNICAL FEASIBILITY

- 1 Today**
Station data + Maps + Vehicle data
- 2 MVP**
Route optimization + Status + Queue estimation
- 3 Next**
Operator integrations + Booking
- 4 Scale**
Payments + Predictive intelligence + Fleet/OEM APIs

BUSINESS MODEL



Charging transaction commission



Reservation fees



B2B operator analytics



Fleet subscriptions & OEM / API integrations

We don't need to own charging stations — we become the intelligence & experience layer connecting users to them.

From charging app to EV mobility intelligence layer

SCALE PATH

- MVP
- Pilot with selected stations
- Multi-network integration
- Fleet & OEM partnerships
- **National EV Intelligence Platform**



We don't just find chargers. We make charging predictable.